

# System for Sign Language to Speech Conversion

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**Abstract**—Communication is the very essence of human life as it helps human beings to communicate with one another. It helps build relationships by making sharing of experiences, thoughts, and learnings a possibility. Speech communication is the easiest way for communication amongst all the various means for interactive communications. However, for the masses with speech impairments, it is extremely challenging to communicate with everyone else. These people make use of sign language for communication with normal people, but the sign language is not known by many people. Whenever a speech-impaired person communicates with normal person, a communication gap is created among which happens to be hard to fill. Hence it is proposed to develop a system that is capable of converting sign language to speech, with an objective to assist the specially abled people communicate with the world seamlessly.

**Keywords** - Flex Sensors, Arduino, Sign Language Translation, Gesture Recognition

## I. INTRODUCTION

In their everyday life the specially abled people from the deaf and mute community have a necessity for interaction with normal people for their everyday set of activities. Across many parts of the world these people make use of sign language which is one of effective way they feel to communicate with people around them. However, to be able to understand the sign language, it is necessary to undergo a special training. It is primarily used by deaf people or those with difficulty in hearing and the mute community for the purpose of communication with others. Each hand movement and expression have a specific meaning, like words in spoken language. Typically, the sign language is dependent on visual communication, using hand movements and gestures. The hand gesticulations are the specific hand movements with an objective to form words based on the gestures made.

The sign language used in different countries is not the same. Several sign languages such as the American Sign Language (ASL), British Sign Language (BSL), Japanese Sign Language (JSL), Indian Sign Language (ISL) and many similar country specific sign languages do exist. An enormous amount of research is being carried out to come up with a simple, compact, affordable, and precise system to convert sign language to speech.

The systems being researched and developed typically consist of use of hand-worn gloves that convert the specific hand gestures to be displayed as text messages and subsequently to speech using Arduino. The development of a simple, practical, and commercially viable system to convert the sign language to speech will helpful to many people. When easy communication becomes a possibility, it gives a new hope and meaning to the lives of the people with speech impairments.

## II. BACKGROUND OVERVIEW

The existing architectures for sign language to speech conversion systems consists of hardware and software parts, the hardware part consists of utilizing flex sensors for every finger on the hand, gloves, bluetooth module and using an accelerometer along with the Arduino as a main controller. The software part comprises of the Arduino (Integrated Development Environment) IDE and the android application [1].

A prototype to identify ASL alphabets and words has been proposed by Aarthi M and Vijayalakshmi P, which uses an Arduino Uno microcontroller to map the readings with the defined threshold values for every finger bending, hand rotation to generate the corresponding text. The HMM based Text-to-Speech synthesizer has been used to translate the text to speech [2]. Using Random Forest Classifier, an automatic sign language recognition system is implemented to recognize and convert the sign alphabets and most common words into text and sound. A glove for hands, circuit with flex sensors, three axis accelerometer and gyroscope are used to detect the gestures. The finger bend data was is provided by the flex sensors on each finger while the accelerometer and trajectories of the hand motion are provided by gyroscope is used to recognize the gestures made [3].

The bluetooth module along with the android smart phone in association with a hand glove to detect the gestures and produce artificial speech has been devised [4][7]. The Support Vector Machine (SVM) algorithm is utilized to categorize the sign language gestures and guess the word [5]. In a similar work, the text message is sent using GSM technology and the speech output using an android application [6]. A simple and cost-effective encoder-based system is developed for deaf-mute people [8]. A detailed analytical study of the various algorithms used in them implementation of Sign Language Recognition (SLR) is developed [9]. An Arduino nano microcontroller based compact prototype is proposed to recognize American Sign Language (ASL) and Indian Sign Language (ISL) also [10]. A glove-based device is developed to perform ASL recognition and convert gestures to messages in text form and further to speech [11].

Similar system is developed making use of just the Arduino and flex sensor yielding a total of 32-bit equivalent options to study the gestures A comparative build of hardware using a 8CHvoice recorder along with speaker instead of android application and bluetooth module is used to get the a similar output [12] [13]. A devise to assist people suffering from mutism is proposed [14] and a similar low power and economical device for sign language to speech is proposed [15].

### III. METHODOLOGY

The overall functioning of the proposed system for gesture recognition is illustrate in Fig. 1.

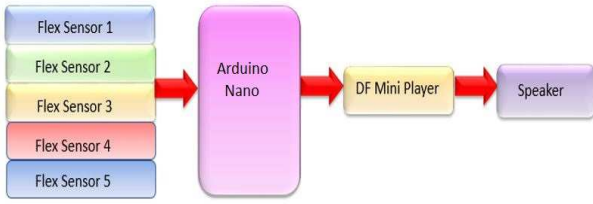


Fig. 1. Block diagram.

#### A. Block Diagram

In the proposed hand-based gesture recognition system, uses a flex sensor to read the resistance and this resistance is mapped to specific phases. The block diagram in Fig. 1. represents the general hierarchy of various working blocks of sign language to speech conversion desired system.

The block diagram represents the high-level flow of information where in, the first step is the mapping of the resistance value to the recognized gesture. The recognized gestures are analysed and interpreted using the data fed into the system, and finally, the interpreted information is converted into spoken words using DF mini player.

#### B. FlowChart

The Fig. 2. represents the control flow of algorithm used. The system iterates continuously tries to identify the hand gesture, subsequently proceeds to translate to text and speech

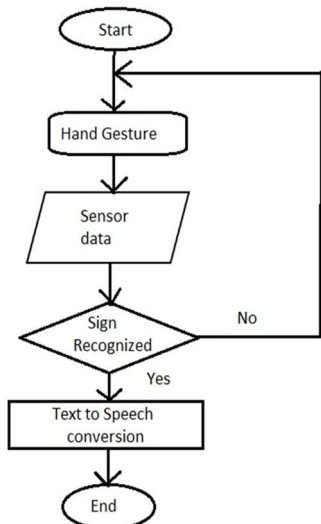


Fig. 2. Flowchart.

The hand gloves that are used in the identification of the finger movements are stitched with flex sensors. An initial voltage is supplied to each flex sensor. Subsequently, a varied voltage-drop as a consequence of the resistance change that

depends of the fingers' bending movement. The values of the voltage variations are time varying in nature. These time varying values are converted to the digital domain. The Arduino Nano with Analog to Digital Converter (ADC) is used to achieve this. The converted digital data is seen to be helpful in further signal processing.

The obtained digital output is compared with the already loaded values of sensor in the system for different phases. Whenever the digital output matches with already existing values, then the hand gestures are recognised. The output is displayed on the Liquid Crystal Display (LCD) which will help a normal person to read and understand. Further the same output is given to the DF miniplayer module to render the speech output. Thus, the system proposes to give a visual display along with an audio rendition of the converted sign language.

### IV. HARDWARE IMPLEMENTATION

The hardware implementation for sign language to speech conversion involved the integration of several hardware components. A sensor gloves capable of capturing the movements accurately is used. These sensors on the gloves and worn on the hand are capable of accurately capturing various sign language gestures. The data captured from all the connected sensors is provided to Arduino Nano microcontroller. To be able to assist further processing the analog values are converted to digital data. The obtained digital data is compared to the previously stored database of the Arduino Nano IDE. Appropriate library functions are utilized to construct a database suitable to the sign language. The gesture made is recognized as soon as the data values from the input values match with the stored values in the database.

The Arduino Nano microcontroller receives the sensor data and executes the necessary computations to generate the appropriate output, which is fed as input to the DF mini player module and the Liquid Crystal Display.

Once the sign language gestures are identified the Arduino Nano generates the corresponding text message to display together with the speech output. The speech synthesis is done using the DF mini player to generate the corresponding speech output. The speech output is rendered using a simple 8 Ohm speaker.

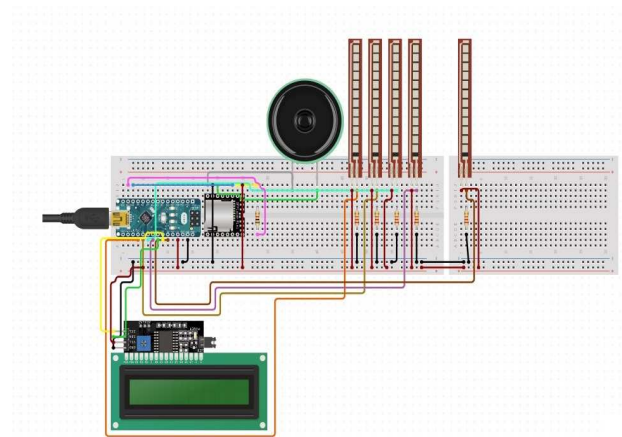


Fig. 3. Image representation of the circuit implemented

The image representation of the implemented circuit is shown in Fig. 3. It shows the five flex sensors connections made to the Arduino Nano microcontroller including the LCD and speaker connections are shown. The 2.2-inch resistive flex sensor used is shown in Fig.4.



Fig. 4. 2.2-inch resistive flex sensor.

The voltage divider circuit image with its connection to the 2.2-inch resistive flex sensor is illustrated in Fig. 5.

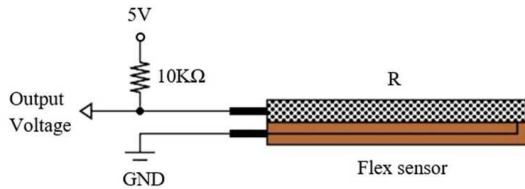


Fig. 5. Flex Sens or Connection

The five flex sensors, one each for every finger are stitched on to the gloves. This forms the main aspect and hence each finger has a separate sensor. The converted text message is displayed using a 16X2 Liquid Crystal Display. An I2C interface for communication is made use for establishing a used for the purpose of connecting with the Arduino. The DF Mini MP3 Player for Arduino is used, which is a small and cost effective MP3 module with an simplified output delivered directly to the speaker. A simple 8 Ohm / 0.25-watt Speaker is used to render the speech signal. The image of the completely implemented circuit including the sensors mounted on the gloves is shown in the Fig. 6.

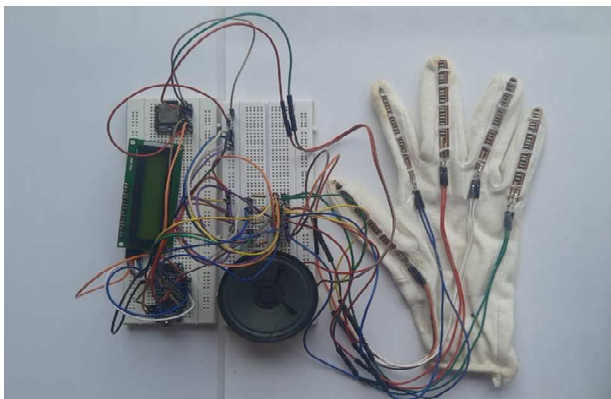


Fig. 6. Image of Implemented Circuit

## V. RESULTS AND DISCUSSION

During the process of implementation, all flex sensors are connected to Arduino Nano board by jumper wires. Once the connections are made, then Arduino takes inputs from these sensors, these values are kept in the database. Using the above system, gestures made are captured are evaluated according to ASL (American Sign Language) and converted to corresponding texts which are displayed on LCD, these texts are converted to speech using DF Player mini and given out as output by speaker.

The designed circuit is tested with simple hand gestures. For instance, the output displaying “thank you” on the Liquid crystal Display for the gesture made is shown in the Fig. 7. The output displaying “please help me” on the Liquid crystal Display for the gesture made is shown in the Fig. 8.

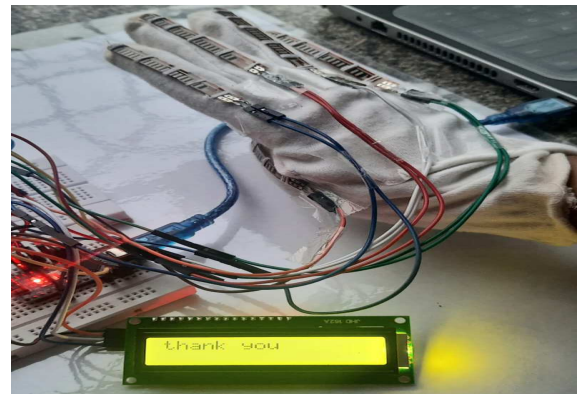


Fig. 7. Output Displaying “thank you”

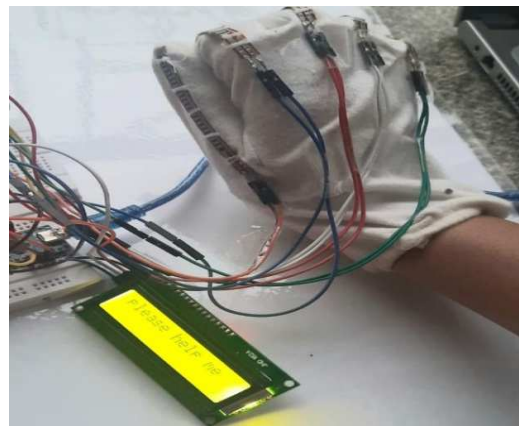


Fig. 8. Output Displaying “please help me”

The hand gloves are made light weight and the circuitry consumes low power. The messages displayed on the screen will help the deaf people to read the message and the sound through the speaker will help the blind people to listen to the message.



can be extended to provide home automation on finger tips. The system may be paired up with fitness sensor to monitor health of the individual.

sketch\_jun24a\_proj1\_8mp3.ino

```

77   myDFPlayer.play(3);
78   delay(1000);
79 }
80 if(((fileADC2>170)&&(fileADC2<190)))
81 {
82   Serial.println("thank you ");
83   lcd.setCursor(0,0);
84   lcd.print("thank you");
85   myDFPlayer.play(4);
86   delay(1000);

```

Output Serial Monitor x

Message (Enter to send message to 'Arduino Nano' on 'COM5')

|                  |               |               |               |               |
|------------------|---------------|---------------|---------------|---------------|
| filex1 186.00    | filex2 212.00 | filex3 150.00 | filex4 220.00 | filex5 330.00 |
| filex1 181.00    | filex2 212.00 | filex3 233.00 | filex4 227.00 | filex5 346.00 |
| filex1 183.00    | filex2 211.00 | filex3 233.00 | filex4 124.00 | filex5 291.00 |
| nice to meet you |               |               |               |               |
| filex1 183.00    | filex2 212.00 | filex3 237.00 | filex4 127.00 | filex5 274.00 |
| nice to meet you |               |               |               |               |
| filex1 206.00    | filex2 215.00 | filex3 243.00 | filex4 213.00 | filex5 288.00 |
| filex1 189.00    | filex2 212.00 | filex3 231.00 | filex4 212.00 | filex5 312.00 |
| filex1 183.00    | filex2 212.00 | filex3 235.00 | filex4 213.00 | filex5 309.00 |
| filex1 185.00    | filex2 214.00 | filex3 236.00 | filex4 158.00 | filex5 269.00 |
| please help me   |               |               |               |               |
| filex1 200.00    | filex2 223.00 | filex3 227.00 | filex4 106.00 | filex5 250.00 |
| nice to meet you |               |               |               |               |
| filex1 192.00    | filex2 214.00 | filex3 236.00 | filex4 192.00 | filex5 249.00 |
| filex1 192.00    | filex2 212.00 | filex3 233.00 | filex4 200.00 | filex5 279.00 |
| filex1 194.00    | filex2 212.00 | filex3 239.00 | filex4 207.00 | filex5 300.00 |
| filex1 194.00    | filex2 212.00 | filex3 234.00 | filex4 207.00 | filex5 263.00 |

```

out Serial Monitor X
sage (Enter to send message to 'Arduino Nano' on 'COM5')
flex1 flex2 flex3 flex4 flex5
#1 186.00 flex2 212.00 flex3 150.00 flex4 220.00 flex5 330.00
#1 181.00 flex2 212.00 flex3 233.00 flex4 227.00 flex5 346.00
#1 183.00 flex2 211.00 flex3 233.00 flex4 124.00 flex5 291.00
# to meet you
#1 183.00 flex2 212.00 flex3 237.00 flex4 127.00 flex5 274.00
# to meet you
#1 206.00 flex2 215.00 flex3 243.00 flex4 213.00 flex5 288.00
#1 188.00 flex2 211.00 flex3 231.00 flex4 212.00 flex5 312.00
#1 183.00 flex2 212.00 flex3 235.00 flex4 233.00 flex5 309.00
#1 186.00 flex2 214.00 flex3 236.00 flex4 158.00 flex5 269.00
use help me
#1 200.00 flex2 223.00 flex3 227.00 flex4 106.00 flex5 250.00
# to meet you
#1 192.00 flex2 214.00 flex3 236.00 flex4 192.00 flex5 249.00
#1 192.00 flex2 212.00 flex3 233.00 flex4 200.00 flex5 279.00
#1 194.00 flex2 212.00 flex3 233.00 flex4 207.00 flex5 300.00
#1 194.00 flex2 212.00 flex3 234.00 flex4 207.00 flex5 263.00

```

The system for sign language to speech conversion helps people from the deaf and dumb community to overcome the barriers in communication. The study of previous and related works reveals the difficulties, limitations, and advantages of developing a gesture-based communication system. The implemented sign language to speech conversion system is simple and compact and majorly uses the library functions of the Arduino IDE. The implemented system is unique due to its ability to acquire the bending values and the related speech output very quickly. It makes use of few components and hence its cost is less compared to other similar gesture recognition systems. Additional number of short flex sensors may be deployed to recognize more gestures. This system